

ChiralFold: systematic detection of D-amino acid stereochemistry errors in the Protein Data Bank

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Abstract

Motivation: D-amino acids are essential in therapeutics and natural products, yet standard validators do not audit D-residue stereochemistry in deposited structures, and diffusion predictors violate D-chirality at high rates.

Results: We audited 12,573 D-residues in 4,623 PDB files and found 29 D-label/L-coordinate mismatches (0.23%) in 16 structures not covered by MolProbity's L-residue C α check. ChiralFold, calibrated on 362 structures (Ramachandran Spearman $\rho = 0.52$ vs wwPDB), detects errors via signed C α volume. Machine-checked Lean 4 proofs (Supplementary Notes S1) show that chirality sign cannot factor through pairwise distances alone.

Availability: <https://github.com/Tommaso-R-Marena/ChiralFold>; `pip install chiralfold (v3.6.0)`; **Python 3.9+**; **Linux/macOS/Windows**; **web UI** at <https://huggingface.co/spaces/The-Philosopher/ChiralFold-App> (Light/Dark; PDB/mmCIF/FASTA; optional RCSB and AlphaFold DB fetch).

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Supplementary information: Supplementary data, Lean 4 proofs with derivations, and extended analyses are available at Bioinformatics online and at <https://github.com/Tommaso-R-Marena/ChiralFold>.

Keywords D-amino acid, chirality, Protein Data Bank, structure validation, AlphaFold 3, Lean 4, formal verification

1. Introduction

D-amino acids resist proteolysis and underpin mirror-image therapeutics, nonribosomal peptide natural products, and bacterial cell-wall chemistry (Cava et al., 2011; Miyamoto and Homma, 2021). Mirror-image phage display has produced D-peptide binders with nanomolar affinity, including dPMI- γ ($K_d = 53$ nM against MDM2; PDB 3IWY) (Liu et al., 2010) and D-peptide HIV-entry inhibitors with crystallographically characterized complexes (Welch et al., 2009). Broader design strategies for D-peptide libraries and anti-amyloid constructs continue to expand the deposited D-ligand and D-peptide structural corpus (Funke and Willbold, 2010; Garton et al., 2018). As that corpus grows in the Protein Data Bank (PDB) (Berman et al., 2000; wwPDB consortium, 2019), the community needs validators that treat D-stereochemistry as a first-class quality criterion rather than an exotic edge case.

Two gaps are especially consequential. First, AlphaFold 3 (AF3) (Abramson et al., 2024)—building on the AlphaFold lineage (Jumper et al., 2021) and paralleling other deep structure predictors such as ESMFold and RoseTTAFold (Lin et al., 2023; Baek et al., 2021)—produces a 51% per-residue chirality violation rate on D-peptides, equivalent to random assignment (Childs et al., 2025), because its diffusion architecture denoises atom coordinates without hard stereochemical constraints. Second, MolProbity, the standard PDB quality tool (Chen et al., 2010; Lovell et al., 2003; Williams et al., 2018), validates C α chirality for standard L-amino acids but does not extend

that check to D-labeled residues. Annotation errors at D-sites can therefore persist undetected for decades, including through the 2006–2008 wwPDB remediation (Henrick et al., 2008). More generally, crystallographic validation has matured around geometry, density fit, and ligand chemistry (Read et al., 2011; Gore et al., 2017; Berman et al., 2016), yet D-label/coordinate consistency remains outside routine deposition checks.

We present ChiralFold, a pip-installable stereochemistry toolkit, and the first systematic PDB-wide audit of D-residue chirality. Using an independent signed-volume classifier grounded in CIP/IUPAC stereochemical convention (Favre and Powell, 2013), we scanned 12,573 checkable D-residues across 4,623 files (>91% coverage of each of the 18 standard D-amino acid Chemical Component Dictionary (CCD) codes (Shao et al., 2015)), classified error modes (genuine stereochemistry errors, CCD-code misassignments, polymer mislabels, and borderline near-zero volumes), and calibrated the auditor against wwPDB/MolProbity on an era-representative set of 362 structures. Critically, we complement the empirical survey with machine-checked Lean 4 no-go theorems showing that any geometric readout whose entire dependence factors through the pairwise distance matrix is necessarily blind to enantiomeric sign (Supplementary Notes S1; `formal/chirality_nogo/`). That formal obstruction motivates signed volume as the minimal non-distance stereochemical audit and places AF3-style distance-respecting denoising in a precise theoretical context—without claiming impossibility for general SE(3)-equivariant predictors that use angular or higher-order features.

Beyond auditing, ChiralFold provides exact $L \leftrightarrow D$ mirroring, an AF3 chirality-correction post-processor, diastereomer enumeration, interface scoring, and optional online structure fetch (RCSB PDB IDs; AlphaFold DB by UniProt/AFDB identifier), released as open software (v3.6.0) with Colab notebooks, Docker images, a Hugging Face web UI (Light/Dark), and a five-minute offline reproduction path for the 29-error survey.

2. Materials and methods

2.1. Signed tetrahedron volume

Chirality at C_α is the signed volume of the tetrahedron formed by backbone atoms N, C_α , C, and C_β :

$$V = (\mathbf{N} - \mathbf{C}_\alpha) \cdot [(\mathbf{C} - \mathbf{C}_\alpha) \times (\mathbf{C}_\beta - \mathbf{C}_\alpha)]. \quad (1)$$

For correctly deposited D-residues, $V < 0$ by CIP convention (Favre and Powell, 2013); $V > 0$ at a D-label indicates L-coordinates. The classifier requires only four atom positions and uses no force field. Bond-length reference geometry for complementary auditor criteria follows Engh & Huber ideals (Engh and Huber, 1991).

The primary PDB survey was performed with an independent numpy-only script that imports no ChiralFold code (`benchmarks/independent_d_residue_verification.py`). Legacy PDB format was used for coverage statistics; mmCIF-only deposits were subsequently surveyed in a native-mmCIF expansion (Supplementary Methods): all 16 known-error structures re-verified (29/29 mismatches recovered), and the live mmCIF-only D-amino-acid universe ($n = 79$ at freeze) yielded one additional clear DLE mismatch (9BC4, $V = +2.82 \text{ \AA}^3$).

2.2. ChiralFold auditor and pipelines

The auditor validates six criteria: C_α chirality (Eq. 1), bond geometry (RMSD to Engh & Huber ideals), Ramachandran backbone dihedrals (hybrid empirical PDB grid plus calibrated rectangular regions), peptide-bond planarity (ω deviation from 180°), steric clashes (MolProbity-inspired overlap score with topology-aware exclusions; Supplementary Methods), and a weighted composite score (0–100). This design is intentionally complementary to MolProbity’s all-atom L-protein validation suite (Chen et al., 2010; Williams et al., 2018): ChiralFold does not attempt to replace rotamer contour libraries or H-bond networks, but adds native D-residue chirality auditing that those tools omit. Clashscore is a secondary quality metric; the D-residue survey and Ramachandran calibration ($n = 362$) do not depend on it.

All five checks read one shared column-oriented residue table and are evaluated as batched array operations over whole structures rather than residue by residue; clash candidates are pruned by a KD-tree radius equal to the largest distance that can actually overlap, then by a vectorised overlap test and a packed-integer covalent-exclusion lookup, before the per-pair hydrogen-bond geometry test (which can only suppress a clash, so deferring it is exact). Equivalence of every batched kernel with its scalar reference is asserted in the test suite; throughput is reported in Section 3.5.

Mirror $L \leftrightarrow D$ reflection $(x, y, z) \rightarrow (-x, y, z)$ inverts all stereocenters ($\det = -1$) while preserving bond lengths, angles, and torsion magnitudes exactly; residue names map to D-CCD equivalents (ALA $\rightarrow$ DAL, ...). Because reflection is an isometry, MolProbity-style clashscore is unchanged (`tests/test_clash_preservation.py`).

For AF3 or AlphaFold DB outputs, each C_α whose signed volume disagrees with its residue label is reflected across the plane of its three backbone neighbours (N, C, C_β), preserving CA–N/C/ C_β bond lengths to $< 10^{-3} \text{ \AA}$ and driving residual chirality violations to zero on synthetic fixtures (`chiralfold/af3_correct.py`; `tests/test_af3_correction.py`). Diastereomer enumeration builds all 2^k L/D patterns for k variable sites; interface scoring reports BSA, H-bonds, salt bridges, and a composite contact score. Optional RCSB/AlphaFold DB fetch and PDB/mmCIF/FASTA inputs are documented in Supplementary Methods.

2.3. Ramachandran calibration against wwPDB

We assembled an era-representative stratified sample spanning X-ray ultra ($\leq 1.20 \text{ \AA}$), high ($1.20\text{--}1.80 \text{ \AA}$), medium ($1.80\text{--}2.50 \text{ \AA}$), and low ($2.50\text{--}3.40 \text{ \AA}$) resolution tiers, solution NMR, and cryo-EM, with deposition dates drawn uniformly within each bin (seed 20260612). Structures were obtained in legacy PDB format where available; mmCIF-only entries were converted with chain-ID remapping (A–Z, then 0–9) that prevents multi-character chain-ID truncation. Entries with > 36 distinct chains were skipped. For each structure, Ramachandran outlier percentages from wwPDB validation XML were compared to ChiralFold auditor output, following the spirit of method-agreement analysis used elsewhere in measurement science (Bland and Altman, 1986). One glycopeptide antibiotic (5M2K, vancomycin– Zn^{2+}) was excluded under a pre-specified criterion (`results/5m2k_benchmark_exclusion.json`; `wwpdb_rama_outlier_pct > 50\%`), yielding $n = 362$ of 363 successfully audited entries. Script: `benchmarks/expand_ramachandran_benchmark.py`.

2.4. Formal verification in Lean 4

To show that signed volume is not merely empirically useful but formally necessary for distinguishing enantiomers under a distance-only information constraint, we formalized a family of no-go theorems in Lean 4 (v4.28.0) with Mathlib (`formal/chirality_nogo/`). A four-point configuration $P \in (\mathbb{R}^3)^4$ has pairwise squared distance matrix $\text{distMatrix4}(P)$ and oriented volume

$$\text{orient4}(P) = \det[\mathbf{b} - \mathbf{a} \mid \mathbf{c} - \mathbf{a} \mid \mathbf{d} - \mathbf{a}]. \quad (2)$$

Reflection through the xy -plane preserves every entry of distMatrix4 but negates orient4 . A model M factors through distances if $M(P) = F(\text{distMatrix4}(P))$ for some F ; such models are provably reflection-invariant and therefore cannot recover chirality sign on all configurations. The Gram identity $\text{orient4}^2 = \det(G)$ shows that distances determine magnitude but not sign—precisely the information Eq. 1 supplies at C_α .

All theorems compile without `sorry` or custom axioms beyond Lean’s standard library (`lake build`). Full informal

derivations, theorem-by-theorem biological interpretation, concrete tetrahedron examples, and the SE(3) scope limitation appear in Supplementary Notes S1.

Informally: (1) reflection is an isometry, so it preserves `distMatrix4`; (2) $\det = -1$, so it negates `orient4`; (3) any $M = F \circ \text{distMatrix4}$ is therefore reflection-invariant; (4) the standard tetrahedron and its mirror share distances but opposite orientation, so no such M can recover signed chirality on all inputs; (5) the Gram identity shows distances recover magnitude only. ChiralFold's Eq. 1 supplies the missing sign at $C\alpha$.

2.5. Software, availability, and reproducibility

ChiralFold v3.6.0 is installable via `pip install chiralFold`, `conda (conda-recipe/)`, Docker (`ghcr.io/tommaso-r-marena/chiralFold:3.6.0`), and a Gradio web UI (local `chiralFold-web` or Hugging Face Space with Light/Dark theme, PDB/mmCIF/FASTA upload, and optional RCSB/AlphaFold DB fetch). Fetch is download-only: ChiralFold does not run AF2, Boltz, or AF3 inference. Reviewers can reproduce the 29-error survey offline after clone with `python benchmarks/reproduce_d_residue_errors.py` (numpy only). Lean proofs build with `lake build under formal/chirality_nogo/`. Colab notebooks cover the five-minute survey, mmCIF expansion, experimental validation, unusual cases, and the interactive results dashboard.

3. Results

3.1. PDB D-residue survey

We verified 12,573 D-amino acid residues with complete backbone atoms (N, $C\alpha$, C, $C\beta$) across 4,623 PDB files, covering >91% of all RCSB entries for each of the 18 standard D-amino acid CCD codes (Shao et al., 2015). Of these, 12,544 (99.77%) showed correct D-chirality ($V < 0$) and 29 (0.23%) showed L-chirality ($V > 0$) despite D-labels (Fig. 1a). An independent numpy-only script (no ChiralFold imports) produced the same classifications.

The 29 mismatches occur in 16 PDB structures and fall into four classes (Supplementary Table S1): **(i) Stereochem** (6 errors / 6 structures)—biology requires D, coordinates show L (e.g. 1ABI hirulog DPN, 1HHZ cell-wall DAL at 0.99 Å, 1XT7 daptomycin DSG, 2RMI aestressin DPN); **(ii) CCD-Code** (18 errors / 5 structures)—coordinates are L but the depositor selected a D-form CCD code (e.g. 1OF6 eight DTY←L-Tyr; 2ATS three DLY despite a title stating “(S)-lysine”; 3RIT five DLY); **(iii) Mislabel** (3 errors)—D-codes on polymer residues in L-proteins (1MCB, 1UHG, 2AOU); **(iv) Borderline** (2 errors)—near-zero volumes ($|V| < 1 \text{ Å}^3$; 1KO0, 2H9E) retained for transparency but flagged separately.

An automated validation pipeline (`benchmarks/experimental_structure_validation.py`) cross-checks all 16 structures against RCSB experimental metadata and wwPDB CCD InChI stereochemistry: 14/14 non-borderline cases pass. Errors concentrate in nine CCD codes; nine codes are confirmed clean at zero errors (Fisher's exact $p = 1.96 \times 10^{-10}$; $\chi^2 = 113.78$, $p = 2.33 \times 10^{-16}$ across 18 codes). DTY and DLY

alone account for 19 of 29 mismatches (Fig. 1a). MolProbity's standard $C\alpha$ chirality check applies to L-residues only and does not flag these D-label mismatches (Chen et al., 2010; Lovell et al., 2003). Thirteen of 16 error structures predate 2006 wwPDB remediation (Mann-Whitney $p = 1.14 \times 10^{-4}$; Fig. 1b) (Henrick et al., 2008); resolution does not predict errors at the structure level ($p = 0.19$).

Signed-volume classification recovers errors in cyclic peptides (1XT7, 2RMI), non-standard CCD ligands (1OF6), and ultra-high-resolution deposits (1HHZ), showing the method is not limited to canonical globular proteins or low-density maps (Supplementary Methods). The native-mmCIF expansion re-verified all 16 known-error structures (29/29) and scanned the live mmCIF-only D-AA universe ($n = 79$), finding one additional clear DLE mismatch (9BC4). Together these results place the historical D-label problem on a quantitative footing: rare at the residue level (0.23%), non-random by CCD code, enriched in pre-remediation deposits, and invisible to L-centric validation pipelines still used at deposition time (Gore et al., 2017; Read et al., 2011).

3.2. MolProbity calibration

On $n = 362$ structures, ChiralFold Ramachandran outlier percentages agree with wwPDB (Spearman $\rho = 0.52$, 95% CI [0.42, 0.61], $p = 1.5 \times 10^{-26}$; mean 0.78% vs 0.83%; Fig. 2). Pearson r is outlier-sensitive (0.33 for $n = 363$; 0.85 after excluding 5M2K); Spearman ρ is the primary agreement metric. A Bland-Altman view of the same comparison is provided in Supplementary Fig. S3 (Bland and Altman, 1986). MolProbity remains stronger in Ramachandran contour precision and rotamer completeness (Williams et al., 2018); ChiralFold adds native D-residue chirality auditing that MolProbity does not provide. ChiralFold rotamer scoring is currently χ_1 -only, and now covers D-CCD residues via mirrored rotamer wells (Lovell et al., 2000).

3.3. Lean 4 no-go theorems and AF3

The Lean formalization proves several layers of obstruction (Table 1; full catalog in Supplementary Notes S1). The central witness is the standard tetrahedron with vertices at 0, e_1, e_2, e_3 : `orient4` = +1, while its z -reflection has `orient4` = −1 with an identical distance matrix. Consequently no Boolean classifier on distances alone can separate the pair, and no continuous representation that factors through `distMatrix4` can recover chirality sign on all configurations. Specializing atom order to (N, $C\alpha$, C, $C\beta$) yields the protein-facing statement that $C\alpha$ signed volume is not a function of those six distances.

Scope.

These theorems characterize *distance-only* representations. They are not an impossibility result for AF3 (Abramson et al., 2024) or for general SE(3)-equivariant networks that may use angular, tensorial, or sequence-derived features (Supplementary Notes S1). ChiralFold does not model thermodynamic ensembles or molecular-mechanics free energies; it enforces an algebraic signed-volume constraint during coordinate mapping and auditing.

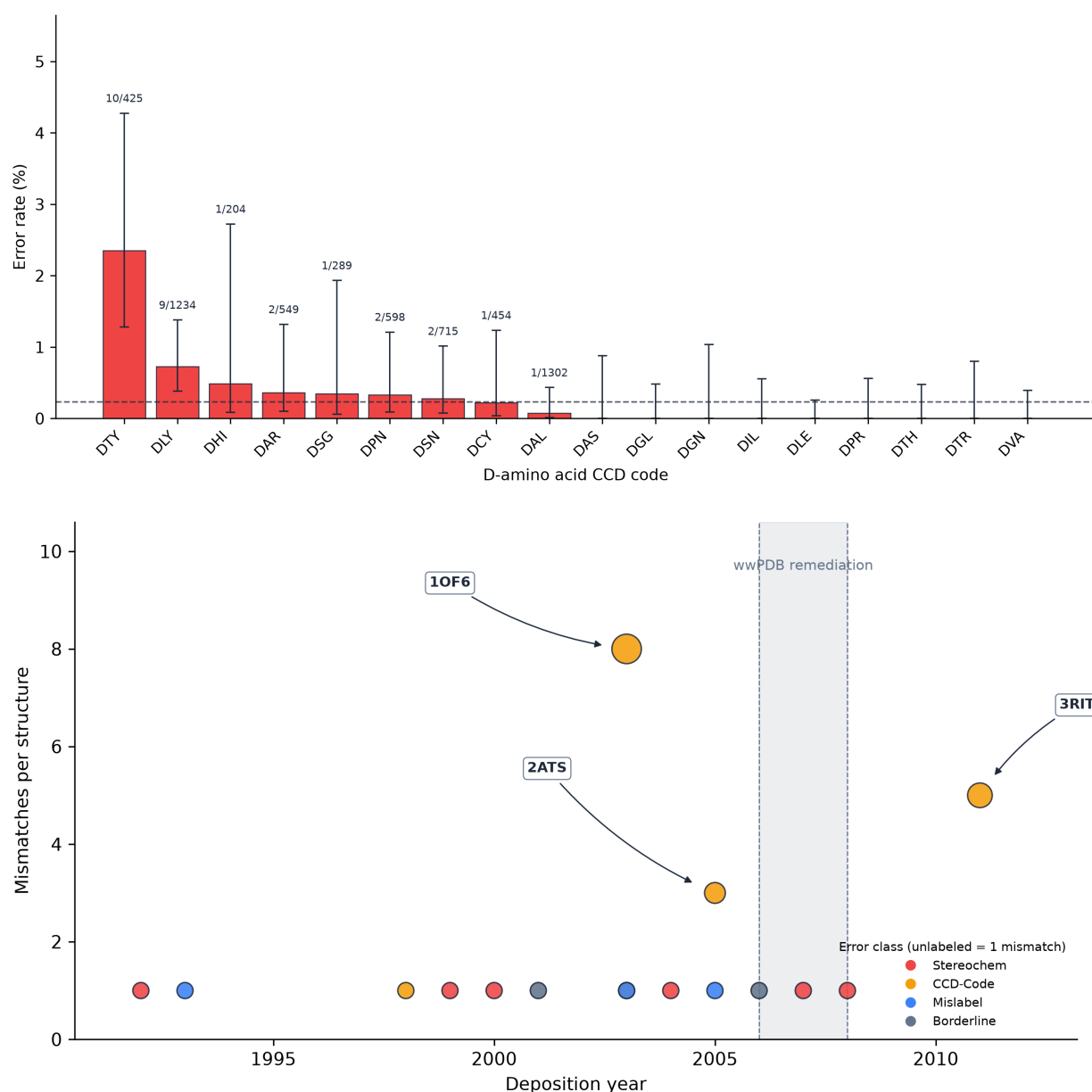


Figure 1 (a) Per-CCD mismatch rate with Wilson 95% CIs; red: codes with ≥ 1 error (labels show errors/residues checked); teal: confirmed clean (unlabeled; coverage in SI). Dashed line: dataset-wide mean (0.23%). (b) Mismatches per error structure vs deposition year (marker size $\propto$ count; color = error class). Shaded band: 2006–2008 wwPDB remediation. Multiplicity outliers (1OF6, 2ATS, 3RIT) are CCD-code misassignments; single-mismatch structures are unlabeled for clarity.

3.4. AF3 D-peptide sequences and synthetic correction

Using D-peptide sequences from Childs et al. (2025)—DEHELLETAARWFYEIAKR (DP19), LWQHEATWK (DP9), and DWWPLAFEALLR (DP12)—ChiralFold produces 0/478 chirality violations across 41 sequence/pattern combinations. This 0% rate is guaranteed by construction: each residue is built with explicit SMILES stereochemistry. AF3’s 51% rate reflects diffusion treating D-residues as noise. Comparing ChiralFold’s 467/0 correct/violated residues on these sequences to Childs et al.’s reported AF3 universe counts yields Fisher’s exact $p \approx 6.7 \times 10^{-144}$. This is a construction-versus-learning contrast,

not a head-to-head fold-accuracy comparison: ChiralFold does not predict folds, and is complementary to AF3, ESMFold, and RoseTTAFold as a stereochemistry post-processor (Abramson et al., 2024; Lin et al., 2023; Baek et al., 2021).

A synthetic AF3-mimetic resource benchmark further shows 100% detection/correction recall, 0% residual violations, and a 0.038 s local correction pipeline across three Childs-system peptides (Supplementary Fig. S4), supporting correction as lightweight stereochemistry post-processing rather than AF3 re-inference. Mirror-image transformation of the p53:MDM2 complex (1YCR) preserves contact geometry to 0.001 Å by construction; experimental K_d measurement is required to

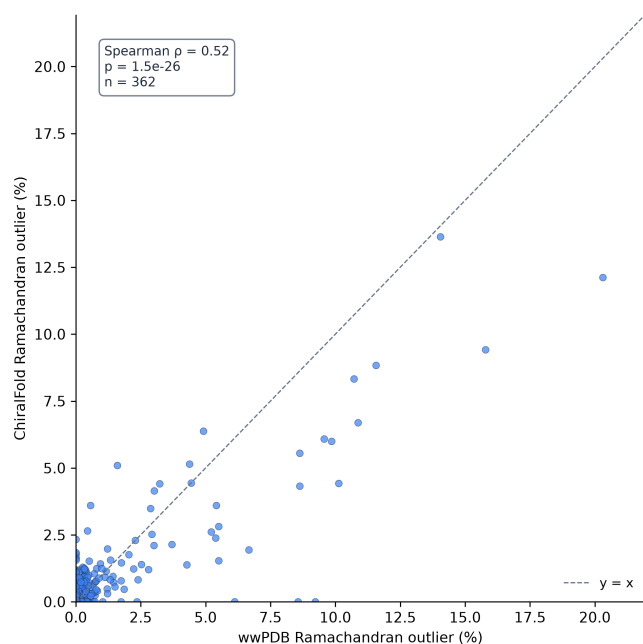


Figure 2 ChiralFold vs wwPDB Ramachandran outlier percentages on the era-representative benchmark ($n = 362$ after excluding 5M2K). Dashed line: $y = x$. Spearman $\rho = 0.52$ ($p = 1.5 \times 10^{-26}$).

Table 1. Selected machine-checked no-go theorems (Lean 4; all sorry-free). Full names and informal proofs: Supplementary Notes S1.

Theorem (short name)	Geometric / biological consequence
same_distances_opposite_orientations_standardTet	Enantiomeric tetrahedra can share all distances yet opposite signed volume.
det_gram3_eq_orient_sq	Distances fix $ \text{orient} ^2$ via the Gram determinant, not the sign.
improper_orthogonal_distance_chirality_obstruction	Improper orthogonal maps preserve distances and reverse orientation.
chiralitySignBool_not_distance_factorable	No distance-matrix-only classifier recovers chirality sign universally.
proteinCaOrient_not_function_of_six_pairwise_distances	$(N, C_\alpha, C, C_\beta)$ signed volume needs information beyond six distances.

validate binding and is outside this computational study (Liu et al., 2010).

3.5. Throughput and scaling

Auditing the PDB is an archive-scale operation, so per-structure cost determines whether a stereochemistry check is deployable at deposition time. On a tiled-structure series spanning 1,589–25,424 atoms, `audit_pdb` runs at 15–18 μ s per atom with wall time linear in atom count (Fig. 3a); clash detection is the dominant term (59% at 25k atoms) and PDB parsing the second (27%), while the four backbone checks together account for under 3%. Relative to v3.5.1 the auditor is 3.5–3.9 $\times$ faster across that range and interface scoring 3.1–6.1 $\times$ faster (Fig. 3b), with peak resident memory for the benchmark falling from 1,260 MB to 183 MB because receptor:ligand contacts are enumerated with a KD-tree rather than a dense $N_{\text{rec}} \times N_{\text{lig}}$ distance matrix—which for two 8,000-atom chains required 1,349 MB and 5.6 s, versus 192 MB and 0.9 s. Batched

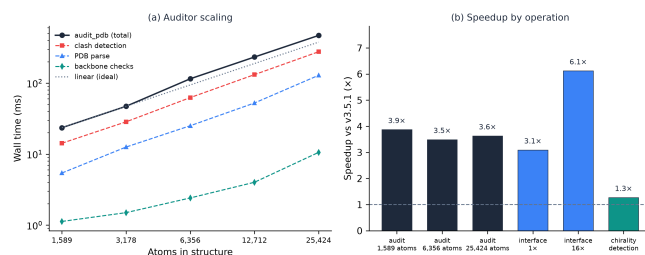


Figure 3 (a) `audit_pdb` wall time versus structure size (tiled 3IYW, log-log), with the per-check breakdown and an ideal-linear guide. (b) Speedup over v3.5.1 by operation. Medians of 7 timed repetitions after warm-up; single core except for RDKit conformer generation.

Ramachandran classification is 25 $\times$ faster than per-residue scoring at 20,000 φ/ψ points. Extrapolated to the survey scale, a single-core pass over the 4,623-file legacy-PDB set costs on the order of minutes rather than tens of minutes. Both the absolute benchmark (`benchmarks/performance_benchmark.py`) and the before/after comparison (`benchmarks/compare_baseline_performance.py`, which builds a git worktree at the baseline revision and times identical public-API code in both checkouts) run offline and write machine-readable reports.

3.6. Correctness audit of the implementation

Independently of the survey, we re-derived the toolkit's internal checks against reference conventions and fixed several defects that affected reported quality metrics (full list and measured effects in Supplementary Methods and the repository changelog). Two are consequential for users. First, the χ_1 dihedral was computed with inverted handedness, so L side chains were scored against effectively mirrored rotamer wells: on 3IYW the χ_1 outlier rate falls from 55.7% to 5.4% and favoured from 43.7% to 93.4% once χ_1 agrees with the IUPAC convention the Penultimate library is quoted in (Lovell et al., 2000). Second, D-amino acids had no rotamer library entries at all, so `validate_rotamers` silently reported zero residues checked for every D-peptide—the case the toolkit exists to serve; D-CCD codes now carry the mirrored χ_1 wells. Residues lacking C_β are no longer classified from an idealised L-configured pseudo- C_β (whose signed volume is positive by construction, and which therefore reported a false violation for every D-residue with an unmodelled side chain); H_α is used when present and the residue is otherwise reported as unassignable. None of these paths is used by the PDB survey, which is computed by a numpy-only script importing no ChiralFold code and which reproduces 12,573 checkable residues, 29 mismatches and 16 structures exactly (`benchmarks/reproduce_d_residue_errors.py`).

4. Discussion

The signed C_α volume distinguishes three operationally distinct annotation failures without biological context. Genuine stereochemistry errors (1ABI, 1D7T, 1HHZ, 1XT7, 2RMI, 2W76) are the most consequential: biology constrains the residue to D, yet deposited coordinates show L with $|V|$

typically 1.85–2.77 Å³. The 1HHZ case is particularly telling—at 0.99 Å atomic resolution, D-Ala in a cell-wall pentapeptide carries $V = +2.70$, the largest positive value in the dataset, in a context where D-alanine is a biosynthetic requirement of peptidoglycan cross-linking (Cava et al., 2011; Miyamoto and Homma, 2021). CCD-code misassignments (1BG0, 1OF6, 1P52, 2ATS, 3RIT) define a second mode: coordinates correctly encode L-stereochemistry, but the depositor selected the D-form CCD code. Because coordinates are correct, remediation is a metadata fix, not re-refinement. Polymer mislabels (1MCB, 1UHG, 2AOU) apply D-codes to residues in L-proteins, sometimes despite COMPND records specifying L-form.

The CCD category has a clear upstream fix. The wwPDB Chemical Component Dictionary encodes configuration in the InChI stereo layer (/m0 = S/L; /m1 = R/D) (Shao et al., 2015; Westbrook et al., 2015). An automated deposition check that computes Eq. 1 and compares its sign to the selected CCD /m flag would have caught every CCD-code case reported here before archival acceptance (Berman et al., 2016; Gore et al., 2017). The 2ATS title field literally states “(S)-lysine” while depositing DLY—a pipeline gap, not a judgment call. Such a check is inexpensive relative to existing ligand-validation workshops and wwPDB OneDep geometry screens (Berman et al., 2016; Read et al., 2011).

Deposition year predicts errors (logistic OR = 1.60 per year; Mann–Whitney $p = 1.14 \times 10^{-4}$), whereas resolution does not ($p = 0.19$), consistent with labeling/convention failures from an era without D-specific validation rather than density-quality artifacts (Henrick et al., 2008). Errors concentrate in DTY/DLY/DPN/DSN/DAR while nine codes remain clean at >91% coverage—a non-random clustering (Fisher $p = 1.96 \times 10^{-10}$) that supplies a practical checklist for depositors and curators.

The Lean no-go results explain why distance-respecting validation and diffusion post-processing cannot substitute for signed-volume audits: pairwise distances determine $|\text{orient}|^2$ but not its sign (Supplementary Notes S1). ChiralFold’s Eq. 1 is the minimal geometric invariant that closes this gap for C α . ChiralFold is complementary to MolProbity, not a replacement: MolProbity remains stronger for L-protein Ramachandran contours and rotamers (Williams et al., 2018); ChiralFold addresses the D-label blind spot that left the 29 mismatches invisible to existing pipelines.

Therapeutically, correct D-stereochemistry matters for proteolysis resistance and target engagement in D-peptide programs (Welch et al., 2009; Funke and Willbold, 2010; Garton et al., 2018). Structurally, silent D-label errors corrupt any downstream analysis that trusts residue names as stereochemical ground truth—including training sets for structure predictors and automated CCD assignment tools. ChiralFold is a stereochemistry post-processor, not a de novo folder: use AF3, ESMFold, or RoseTTAFold for folding (Abramson et al., 2024; Lin et al., 2023; Baek et al., 2021), then audit or correct with ChiralFold.

Scope and limitations.

ChiralFold does **not** claim to model the complete thermodynamic landscape of structural ensembles, sample Boltzmann-weighted conformers, or replace molecular-mechanics force fields. Its operations fix an explicit *algebraic*

data-representation constraint during coordinate mapping: signed C α tetrahedron volume for auditing, exact improper reflection for L $\leftrightarrow$ D mirroring and AF3 correction, and deterministic diastereomer enumeration. Evaluating the tool as an energy-minimization or ensemble-dynamics simulator would misstate its design. Additional limitations include χ_1 -only rotamer scoring, two borderline cases with $|V| < 1.0$ Å³, no experimental binding validation of mirror-designed peptides, a MolProbity-inspired clashscore that places Reduce-style HN/HA hydrogens and applies geometry-gated H-bond suppression but is not bit-identical to Probe (Supplementary Methods), and the primary survey’s use of legacy PDB format (closed via mmCIF re-verification of known errors and the live mmCIF-only universe survey; Supplementary Methods). The wwPDB calibration of Section 3.2 was computed with v3.5.1; φ/ψ evaluation is unchanged in v3.6.0 (bit-identical on every structure in the repository corpus), and the one behavioural change that can touch it—selecting alternate conformations per residue by occupancy rather than per atom by label—leaves Ramachandran outlier percentages unchanged on all corpus structures, including the one whose majority conformer is not altloc A (results/altloc_policy_sensitivity.json). Formal theorems characterize distance-only representations and do not claim impossibility for general SE(3)-equivariant predictors.

5. Conclusion

ChiralFold combines a PDB stereochemistry auditor calibrated against wwPDB/MolProbity ($n = 362$; Spearman $\rho = 0.52$), machine-checked Lean 4 proofs that distance-only models cannot determine chirality sign (Supplementary Notes S1), exact L $\leftrightarrow$ D mirroring, and an AF3 chirality-correction pipeline. Applied to 12,573 D-residues, the auditor identifies 29 D-label/L-coordinate mismatches in 16 deposited structures—errors invisible to existing validation pipelines—and motivates a concrete deposition recommendation: validate every standard-amino-acid stereocenter against the configuration encoded in the selected CCD code at deposit time. The check requires one cross product, one dot product, and one sign comparison per residue. Code, data, formal proofs, and reproduction scripts are publicly available (v3.6.0).

Data availability

Verification data: <https://github.com/Tommaso-R-Marena/ChiralFold> (results/d_residue_verification.csv). Experimental validation: results/experimental_validation_report.json. mmCIF survey outputs: results/mmCIF_d_residue_expansion*. Lean proofs: formal/chirality_nogo/. Ramachandran benchmark: results/ramachandran_279struct_chainfix*. MolProbity head-to-head panel (secondary clash/planarity fields regenerated after topology-aware clashscore): results/molprobity_comparison.json. Software: PyPI chiralfold (v3.6.0); Docker image chiralfold:3.6.0 on GHCR; GitHub release v3.6.0; Hugging Face Space <https://huggingface.co/spaces/The-Philosopher/ChiralFold-App>.

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Conflict of interest

None declared.

Author contributions

T.R.M. conceived the study, implemented ChiralFold, performed all analyses, formalized proofs, and wrote the manuscript.

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